MODULE 43 · FOUNDATIONS - ALL DOMAINS

# Numbers and graphs clinic

Every formula the exam uses on four slides, the traps built into the options, and a method for reading charts quickly, with exhibits to practise on.

DOMAIN Cross-domain  EXAM WEIGHT All domains  MODULE 43 of 49

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# Most exam maths is one formula and one trap.

# Estimating formulas

| MEASURE | FORMULA | MODULE |
|---|---|---|
| Triangular average | (O + M + P) / 3 | 21 |
| PERT, or beta, average | (O + 4M + P) / 6 | 21 |
| PERT standard deviation | (P - O) / 6 | 21 |
| Confidence ranges | +/-1 SD about 68%; +/-2 SD about 95%; +/-3 SD about 99.7% | 21 |
| Velocity forecast | Points remaining / average velocity = iterations | 21 |
| Little's Law | Cycle time = work in progress / throughput | 05 |

ON THE EXAM

Read whether the question says PERT, beta or triangular. If it only gives three points, PERT is the usual default.

# Earned value formulas

| MEASURE | FORMULA | BAD NEWS WHEN |
|---|---|---|
| Cost variance | CV = EV - AC | Negative |
| Schedule variance | SV = EV - PV | Negative |
| Cost performance index | CPI = EV / AC | Below 1 |
| Schedule performance index | SPI = EV / PV | Below 1 |
| To-complete index | TCPI = (BAC - EV) / (BAC - AC) | Above 1, and above CPI |

ON THE EXAM

Every formula starts with EV. Variance subtracts; index divides.

# Forecasting formulas

| MEASURE | FORMULA | USE WHEN |
|---|---|---|
| EAC, typical | BAC / CPI | Current performance will continue |
| EAC, atypical | AC + (BAC - EV) | The variance was a one-off |
| EAC, both | AC + (BAC - EV) / (CPI x SPI) | Schedule pressure also affects cost |
| Estimate to complete | ETC = EAC - AC | How much more will be spent |
| Variance at completion | VAC = BAC - EAC | Negative: forecast overrun |

ON THE EXAM

The story in the question picks the EAC formula: 'continue', 'one-off', or 'both cost and schedule'.

# Everything else

| MEASURE | FORMULA OR RULE | MODULE |
|---|---|---|
| Communication channels | n x (n - 1) / 2 | 14 |
| Expected monetary value | Probability x impact; threats negative | 36 |
| Decision tree | EMV of each branch, including its cost; choose highest | 36 |
| Float | Late start - early start; critical path float is zero | 22 |
| Incentive fee | Seller's share of underrun added to fee; of overrun, removed | 24 |
| Cost baseline and budget | Estimates + contingency; + management reserve | 26 |

ON THE EXAM

Include the project manager in n. Keep EMV signs. Choose the decision branch with the highest EMV.

# Selection rules

| MEASURE | CHOOSE |
| --- | --- |
| Net present value | Highest. Ignore project length: timing is already in it |
| Internal rate of return | Highest |
| Return on investment | Highest |
| Benefit-cost ratio | Highest; above 1 means benefits exceed costs |
| Payback period | Shortest |
| Opportunity cost | The value of the best option not chosen |

ON THE EXAM

Compare like with like. If every option gives NPV, decide on NPV and ignore distracting extra measures.

# The traps in the options

- **Total or increase**: channels after a team grows; the new total, or how many more?
- **Units**: sprints or weeks; thousands or dollars; percent or points.
- **The wrong formula**: triangular for PERT; typical EAC for a one-off.
- **Signs**: an opportunity treated as a threat; a negative variance read as good.
- **Order and share**: buyer and seller shares swapped; TCPI upside down.
- **Intermediate numbers**: an option that is a step on the way, not the answer.

> ON THE EXAM
>
> After you calculate, check your answer is not one of these traps. Several options usually are.

# Calculating under time pressure

- **Write the formula** first, on the whiteboard, before any numbers.
- **Plug in** the numbers, labelled, so you do not swap PV and AC.
- **Estimate** roughly before calculating: is the answer bigger or smaller than the budget?
- **Sanity-check** the result: a CPI below 1 must mean over budget.
- Use the on-screen calculator; spend about a minute; flag and move on if stuck.

> ON THE EXAM
>
> Formula, numbers, estimate, check. A minute each, then move on.

# Reading any chart in five steps

01

### Title and axes

What is measured, in what units, over what?

02

### Legend

Which line or bar is which?

03

### Key point

Find 'today', the end, or the threshold.

04

### Compare

Which line is above which, and is the gap growing?

05

### The question

Read what it asks, then take only that from the chart.

ON THE EXAM

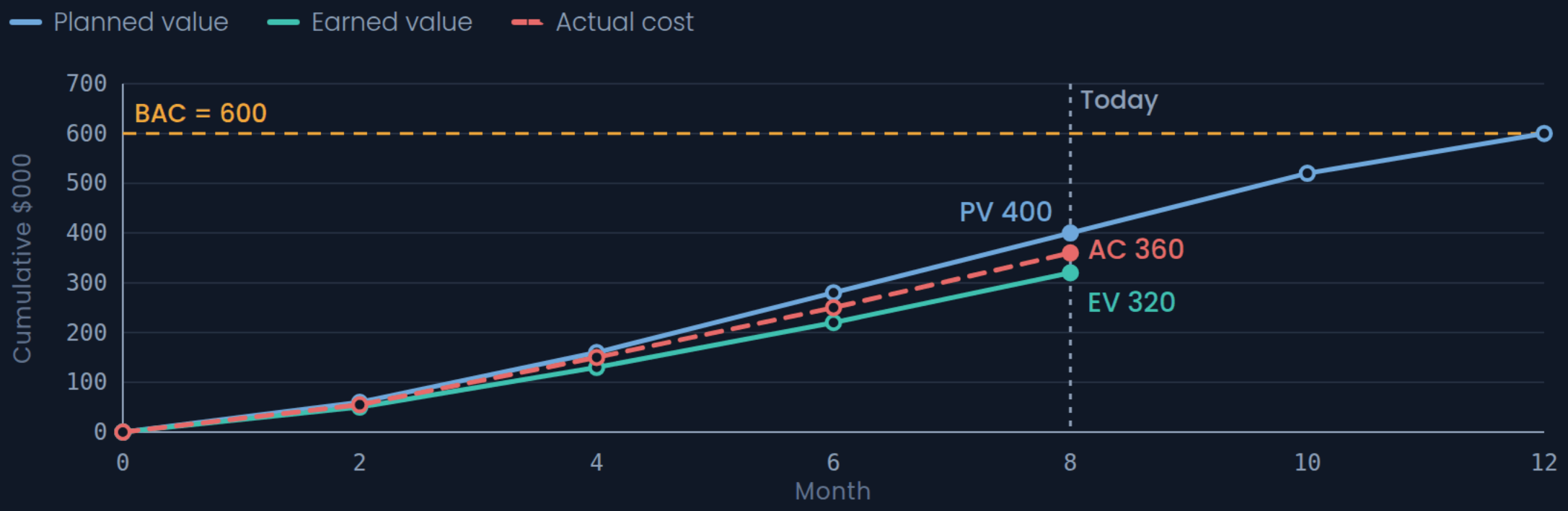
Axes and legend first. Most chart mistakes are reading the wrong line or the wrong unit.

# Chart traps

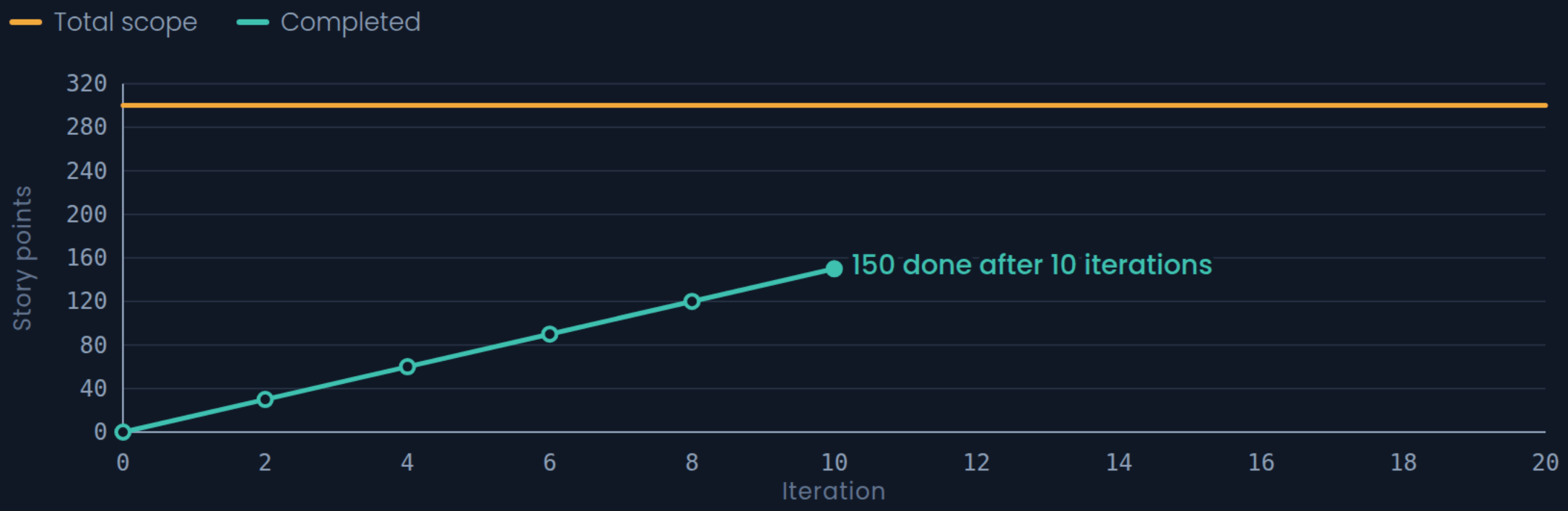
- An axis that does not start at zero makes small differences look large.
- **Cumulative** or **per-period**: an S-curve is cumulative; a velocity bar chart is not.
- Units in the axis title: '$000' means thousands.
- A label near a line is not always that line's value; read the axis.
- When in doubt, read the value at the exact point the question names.

> ON THE EXAM
>
> Check the axis start, the units, and whether the chart is cumulative, before you read a single value.

# Earned value at month 8



# Release burnup



> Refer to exhibit A. At month 8, planned value is $400,000, earned value is $320,000, and actual cost is $360,000.
>
> The sponsor asks for the project's status.

**Based on the exhibit, what is the project's status?**

A Ahead of schedule and under budget.

B Behind schedule but under budget.

C Behind schedule and over budget.

D Ahead of schedule but over budget.

◆ PAUSE AND DECIDE

## Based on the exhibit, what is the project's status?

A Ahead and under.
*EV is below both lines: both indices are below 1.*

B Behind but under.
*Right on schedule; AC is above EV, so over budget.*

C Behind and over.
SPI = 320 / 400 = 0.8; CPI = 320 / 360 = 0.89.

D Ahead but over.
*Right on cost; EV is below PV, so behind.*

THE SIGNAL

On the chart: the green line below both others means behind and over. No arithmetic needed.

> Refer to exhibit A again. The budget at completion is $600,000.
>
> The team expects both the cost and the schedule performance so far to keep affecting the remaining work.

**What is the estimate at completion, to the nearest thousand?**

A $754,000

B $675,000

C $640,000

D $600,000

◆ PAUSE AND DECIDE

## What is the estimate at completion, to the nearest thousand?

**A** $754,000

360 + (600 - 320) / (0.889 x 0.8) = 360 + 394 = 754.

**B** $675,000

*BAC / CPI: the formula for cost performance only.*

**C** $640,000

*AC + (BAC - EV): the formula for a one-off variance.*

**D** $600,000

*The original budget; ignores all variance.*

THE SIGNAL

'Both cost and schedule' picks the CPI x SPI formula. The other two EAC formulas are the distractors.

> Refer to exhibit B. The team has completed 150 of 300 story points in ten iterations, at a steady rate. Scope is not expected to change.
>
> The product owner asks for a forecast.

**At the current rate, how many more iterations are needed to finish the release?**

A 5

B 15

C 20

D 10

◆ PAUSE AND DECIDE

## At the current rate, how many more iterations are needed to finish the release?

A 5

*Would need 30 points per iteration, double the rate shown.*

B 15

*The velocity, not the number of iterations.*

C 20

*The total iterations for the whole release, not the remaining.*

D 10

Velocity 150 / 10 = 15; remaining 150 / 15 = 10.

THE SIGNAL

'How many more' means remaining. Total and velocity both appear as distractors.

> A project team of six people, including the project manager, grows to ten people.
>
> The project manager is updating the communications plan.

**How many communication channels does the team now have in total?**

A 30

B 45

C 4

D 90

◆ PAUSE AND DECIDE

## How many communication channels does the team now have in total?

A 30

*The increase: 45 - 15. The question asks for the total.*

B 45

10 x 9 / 2 = 45.

C 4

*The number of new people.*

D 90

*Forgot to divide by 2.*

THE SIGNAL

This time the question asks for the total. Module 14's scenario asked for the increase. Read the word.

> An activity has a PERT estimate of 20 days. Its optimistic estimate is 14 days and its pessimistic estimate is 32 days.
>
> The sponsor wants a range with about 95 percent confidence.

**What range should the project manager give?**

A 14 to 26 days

B 17 to 23 days

C 11 to 29 days

D 14 to 32 days

◆ PAUSE AND DECIDE

## What range should the project manager give?

**A** 14 to 26 days

SD = (32 - 14) / 6 = 3; 20 +/- 2 x 3 = 14 to 26.

**B** 17 to 23 days

*Plus or minus one SD: only about 68 percent.*

**C** 11 to 29 days

*Plus or minus three SD: about 99.7 percent.*

**D** 14 to 32 days

*The optimistic to pessimistic span, not a confidence range.*

THE SIGNAL

95 percent means two standard deviations. The other ranges are one SD, three SD, and the raw span.

> A Kanban support team has about 30 items in progress at any time. It completes about 10 items a week.
>
> A stakeholder asks how long a typical item takes from start to finish.

**Using Little's Law, what is the average cycle time?**

A 300 weeks

B 0.33 weeks

C 10 weeks

D 3 weeks

◆ PAUSE AND DECIDE

## Using Little's Law, what is the average cycle time?

A 300 weeks

*Multiplied instead of divided.*

B 0.33 weeks

*Divided the wrong way: throughput over WIP.*

C 10 weeks

*That is the throughput per week, not a time.*

D 3 weeks

Cycle time = WIP / throughput = 30 / 10 = 3.

THE SIGNAL

Cycle time = work in progress / throughput. To shorten it, reduce WIP.

Formula, numbers, estimate, check. **Axes and legend first.**